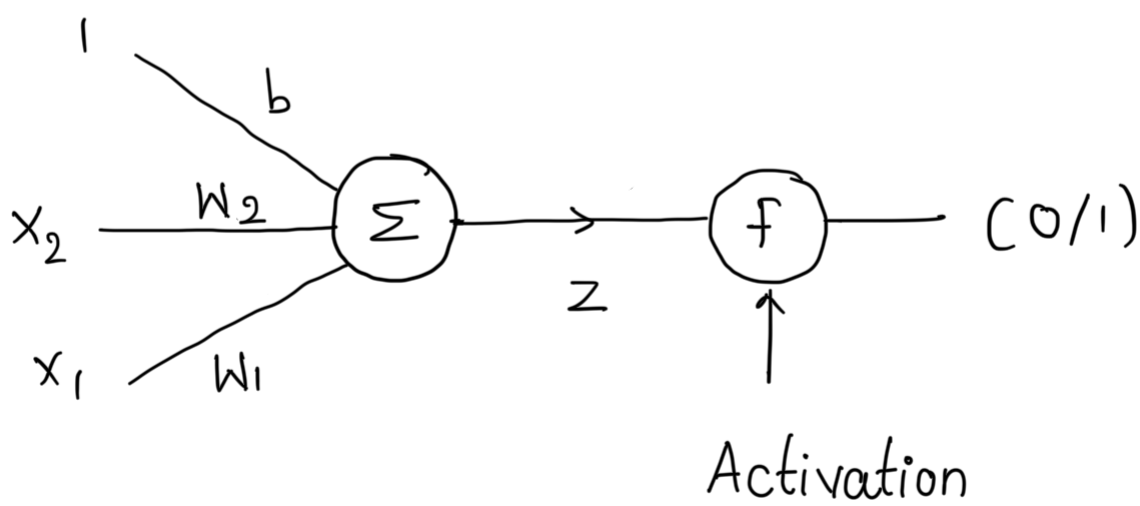
# Perceptron

It is an algorithm which is used in Supervised ML

It is the simplest, Foundational type of ANN used for supervised binary classification, where it maps input data to one of two classes.

It calculates a weighted sum of inputs & applies an activation function, such as a step function, to determine the output. (0/1)



Function

$$z = W_1 x_1 + W_2 x_2 + b$$